TWO OUT OF THREE FOR MORPHISMS OF DISTINGUISHED TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 641-670, following the statement at 630-639.

Editorial note. No omissions: the remaining cases are by rotation (TR2), following the established mirror counting practice.

Allowed standard prerequisites (editorial).

- Representable functor is homological. lemma-representable-homological (same chapter, cited at ~661). The long exact Hom sequence behind the five-lemma argument.
- Five lemma and Yoneda. Homology, Lemma homology-lemma-five-lemma (cited at ~663) and Categories, Lemma categories-lemma-yoneda (cited at ~666). Middle vertical arrow bijective; Yoneda upgrades to b being an isomorphism.

Source reference. Original reference 001P: focused-seeds/geometry/S9-der-third-isomorphism-triangle/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QB: focused-seeds/geometry/S9-der-third-isomorphism-triangle/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let*

$$(a, b, c) : (X, Y, Z, f, g, h) \to (X', Y', Z', f', g', h')$$

be a morphism of distinguished triangles. If two among a, b, c are isomorphisms so is the third.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume that a and c are isomorphisms. For any object W of $\mathcal{D}$ write $H_W(-) = \mathrm{Hom}_{\mathcal{D}}(W, -)$. Then we get a commutative diagram of abelian groups

$$\begin{array}{ccccccccc} H_W(Z[-1]) & \longrightarrow & H_W(X) & \longrightarrow & H_W(Y) & \longrightarrow & H_W(Z) & \longrightarrow & H_W(X[1]) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_W(Z'[-1]) & \longrightarrow & H_W(X') & \longrightarrow & H_W(Y') & \longrightarrow & H_W(Z') & \longrightarrow & H_W(X'[1]) \end{array}$$

By assumption the right two and left two vertical arrows are bijective. As H_W is homological by Lemma 0149 and the five lemma (Homology, Lemma 05QB) it follows that the middle vertical arrow is an isomorphism. Hence by Yoneda's lemma, see Categories, Lemma 001P we see that b is an isomorphism. This implies the other cases by rotating (using TR2). □